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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

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Experiment – 2: Bare-metal Virtualization

Aim: To study and Implement Bare-metal Virtualization using Xen (Open Source tool), HyperV or VMware Esxi.

Objective: After performing the experiment, the students will be able to learn –
- Understand functionalities of OS and Hypervisor
- Working of Hypervisor
- Getting familiar with key concepts of virtualization
- Types of virtualization
- Usefulness of virtualization

Lab objective mapped : ITDO8024.1: To get familiar and implement different types of virtualization techniques.

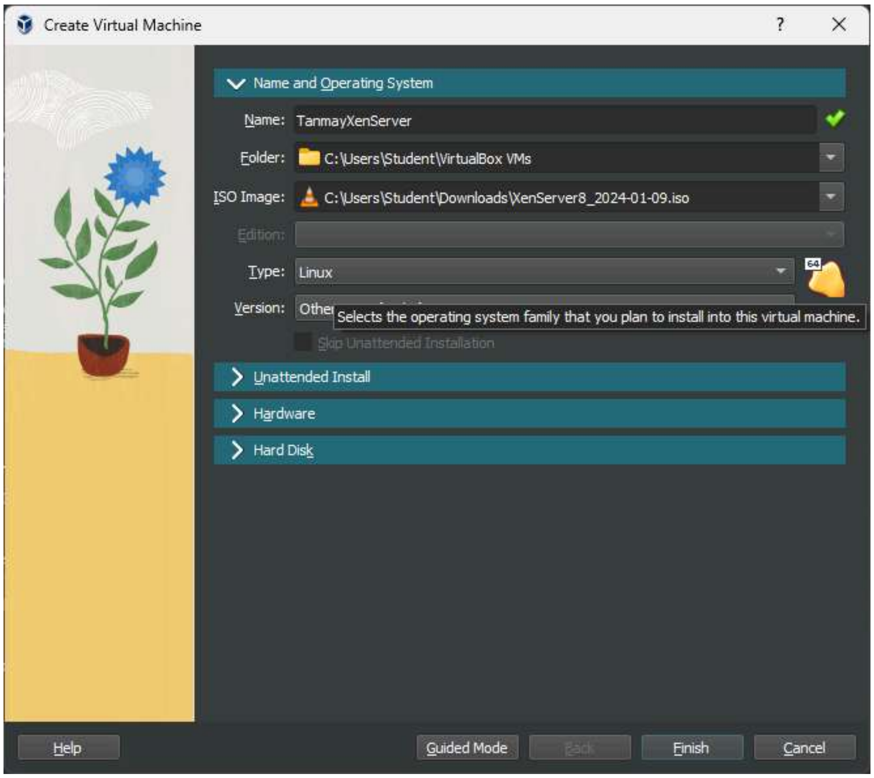
Prerequisite: An AWS account

Requirements:
- XenServer ISO

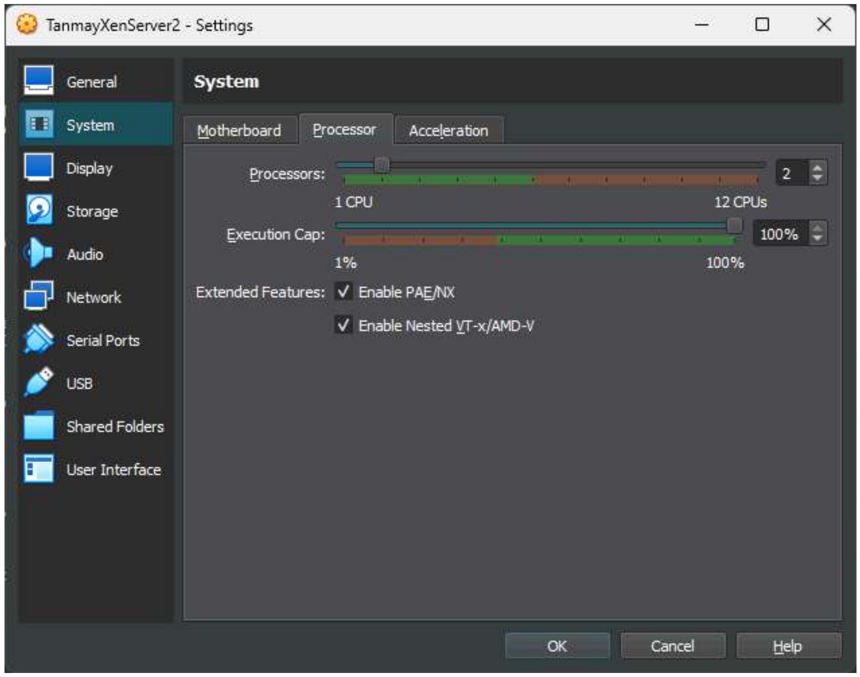
Pre-Experiment Theory:
1. **Hypervisor** (description-To be written in hand))
2. **Bare-metal hypervisor Architecture-**(To be drawn in hand)
3. **Xen Architecture-**(To be drawn in hand)

Procedure (Soft Copy form)
- **Install Xen Server**
  1. Download the Citrix Hypervisor (XenServer) .iso file Download
  2. Create an account to complete the download
  3. Launch Virtualbox
  4. Create a New VM by selecting Machine > New
     i. Name: XenServer
     ii. Machine Folder: C:\VMs
     iii. Type: Linux

iv. Version: Other (64-bit)
v. Memory Size: 8192 MB
vi. Hard disk: Create a virtual hard disk now
5. Click Create
6. On the Create Virtual Hard Disk dialog
i. Name the virtual disk image XenServer.vdi
ii. File size: 500 GB
iii. Hard disk file type: VDI
iv. Storage on physical hard disk: Dynamically Allocated

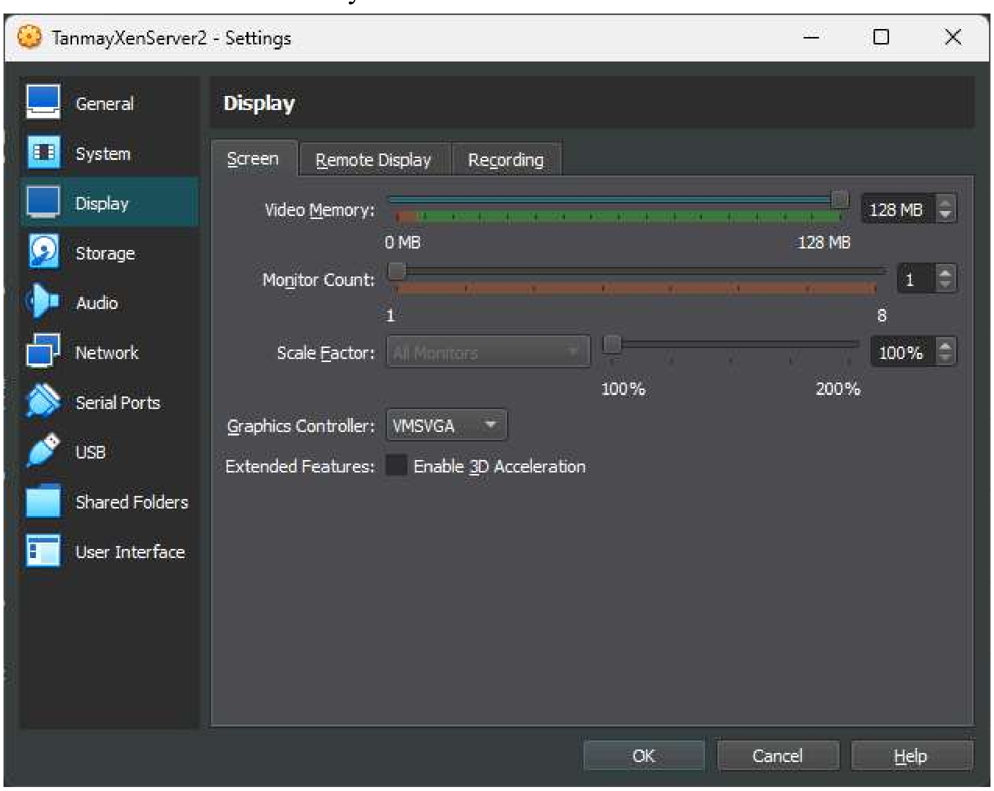


7. Click Create
8. Select the VM and Click Settings
9. Select System > Processor
10. Give the VM at least 2 processors
11. Check the Enable Nested VT-x/AMD-V



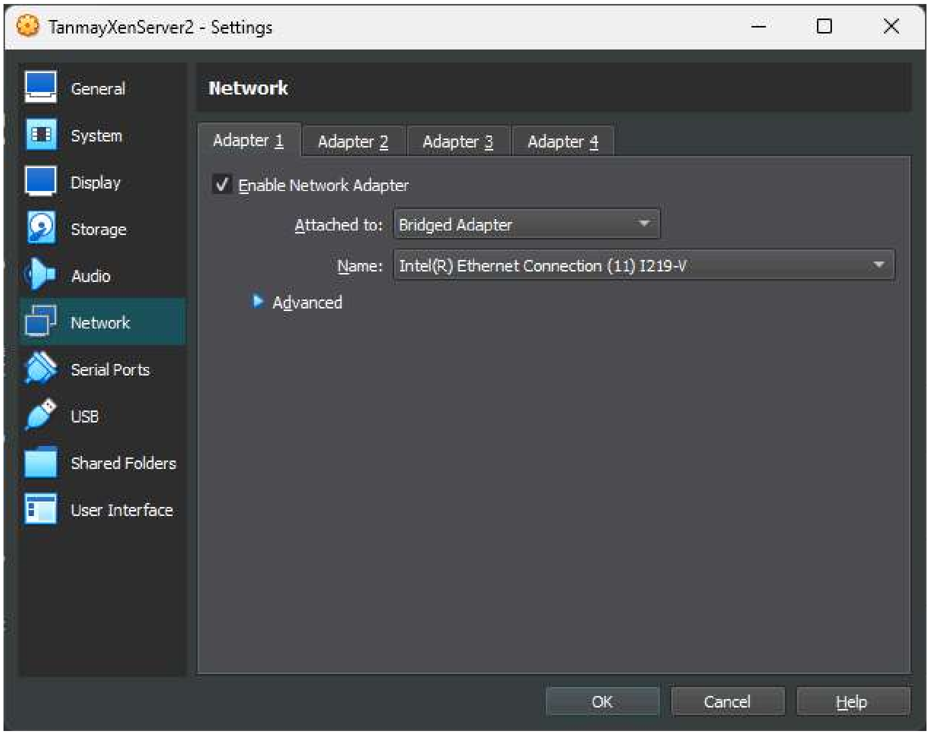
Steps 1-11

1. Select Display
2. Slide the Video Memory to 128 MB



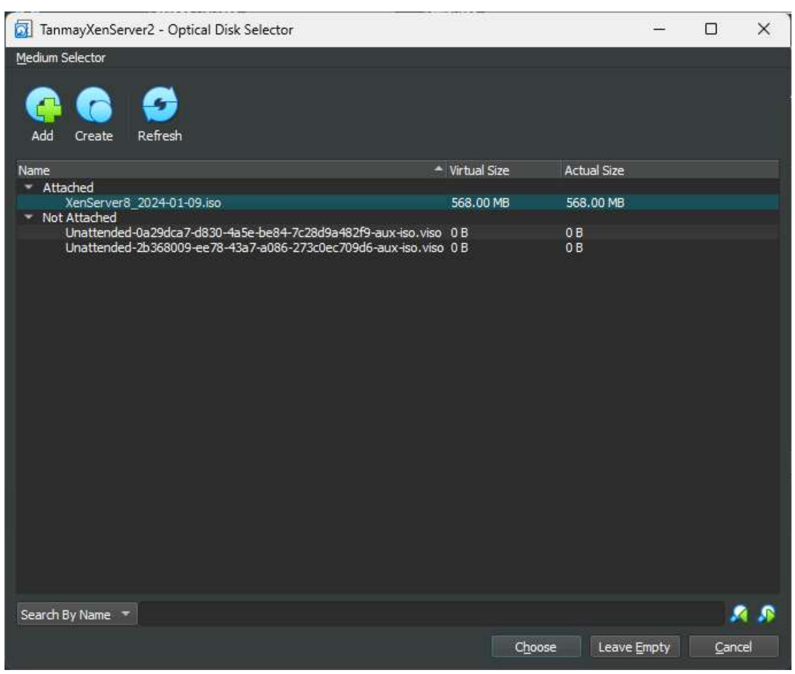
Steps 12-13

14. Select Network
15. Set the attached to dropdown to Bridged Adapter



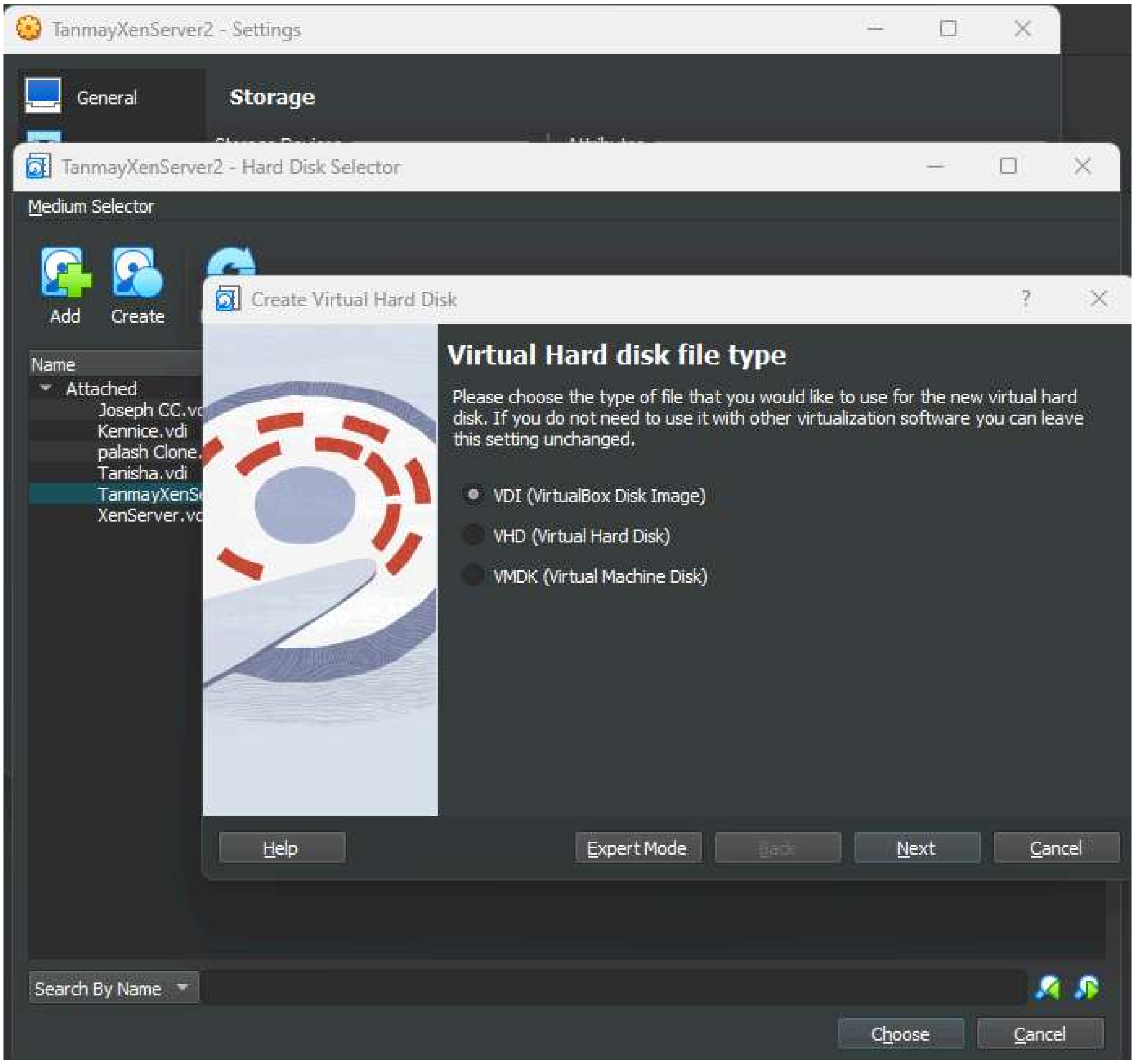
Steps 14-15

5. Select Storage
6. Click on the CD-ROM drive
7. Select the disc dropdown to the right > Choose a virtual optical disc file...
8. Browse to and select the downloaded Citrix Hypervisor (XenServer) .iso file
9. Click OK

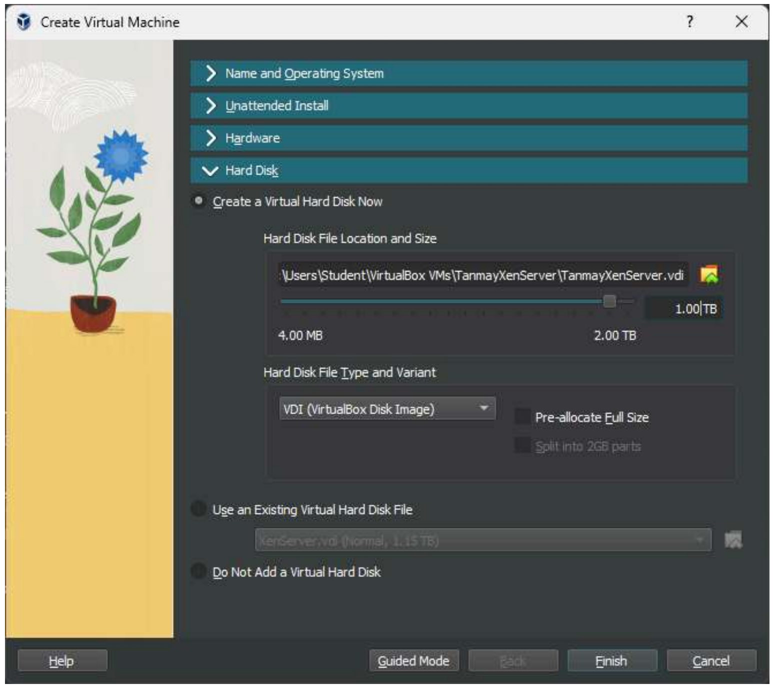


Steps 16-20

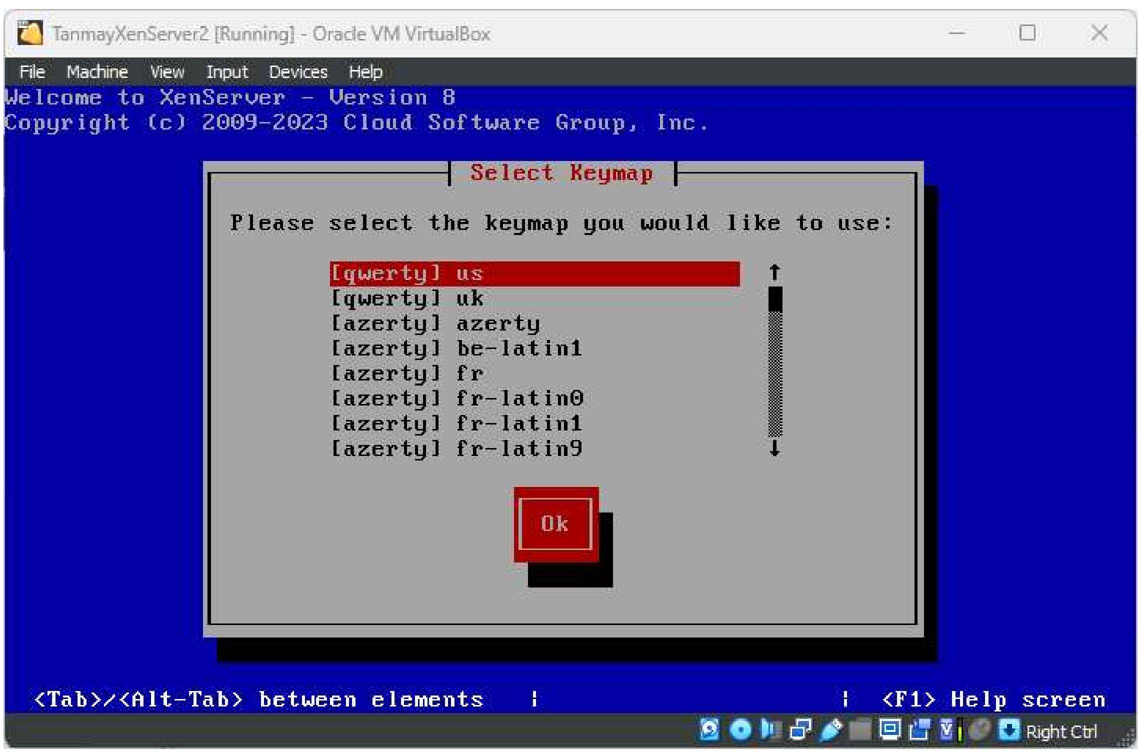
21. Select the hard disk dropdown to the right
11. Click the Create button at the top
12. Change the size to 500 GB > Click Create
13. Click OK to accept the settings



14. Make sure the XenServer VM is selected and click Start > Normal
15. Select a keyboard mapping > Ok
16. Select Ok to begin the installation
28. Select Accept EULA
29. Press Spacebar to select the 500GB VBOX HARDDISK



30. Arrow down and check the Enable thin provisioning box
31. Select Ok



32. Select Local Media > Ok
33. Enter and confirm a root password
34. Configure the network > Ok
35. Specify a hostname > Ok
36. Set the time zone > Ok

37. Select Using NTP > Ok
38. Select Ok at the NTP server setup
39. Select Install Citrix Hypervisor
40. Select No at the supplemental packs installation screen
41. After the installation completes, select Devices > Optical Drives > Remove disk from virtual drive
42. Select Ok to reboot
43. Welcome to Citrix Hypervisor (XenServer) inside VirtualBox

- **Installing Citrix XenCenter**

1. Go back to the download page and download XenCenter Download
2. Install the downloaded .msi
3. Launch Citrix XenCenter
4. Click the Add a Server option in the main window
5. Enter the IP address and root password to the Citrix Hypervisor VM

- **To connect XenCenter to the Citrix Hypervisor server:**

a. Launch XenCenter. The program opens to the Home tab.
b. Click the Add New Server icon.
c. Enter the IP address of the Citrix Hypervisor server in the Server field. Type the root user name and password that you set during Citrix Hypervisor installation. Click Add.
d. The first time you add a host, the Save and Restore Connection State dialog box appears.
e. This dialog enables to set preferences for storing host connection information and automatically restoring host connections.

Post-Experiments Exercise

Attach Followin Screen Shots-

1. Creation of VM inside Oracle Box
2. Any two screen shots of Xen hypervisor installation
3. final working screen of xen hypervisor
4. Enter the IP address and root password to the Citrix Hypervisor VM from xen center

Extended Theory:

1. Difference between bare-metal and hosted hypervisors?(to be written in hand)
2. Bare-metal hypervisor use cases(to be written in hand)

3. Benefits and drawbacks of bare-metal hypervisors(soft copy)
   Benefits of Bare-Metal Hypervisors (Type 1)
   - Bare-metal hypervisors run directly on physical hardware, which provides near-native performance with minimal overhead.
   - They offer better security because there is no host operating system layer.
   - They provide high scalability, making them suitable for enterprise and cloud environments.
   - They support advanced features such as live migration, high availability, and load balancing.

- They improve resource management and isolation between virtual machines.

Drawbacks of Bare-Metal Hypervisors

- They require specialized hardware and technical expertise for installation and management.
- The initial setup cost can be high, especially in enterprise environments.
- They are generally not suitable for small-scale personal use.
- Troubleshooting can be more complex compared to hosted hypervisors.

4. Top bare-metal hypervisor vendors and products(soft copy)

Top Bare-Metal Hypervisor Vendors and Products

1) VMware – VMware ESXi

A widely used enterprise-grade hypervisor known for stability, scalability, and advanced virtualization features.

2) Microsoft – Microsoft Hyper-V (Server/Core)

Integrated with Windows Server, commonly used in enterprise IT infrastructures.

3) Red Hat – Red Hat Virtualization (RHV) (based on KVM)

Enterprise virtualization solution built on the KVM hypervisor.

4) Citrix – Citrix Hypervisor (formerly XenServer)

Based on the Xen hypervisor, used in cloud and enterprise environments.

5) Oracle – Oracle VM Server

Designed for enterprise workloads and Oracle-based applications.

Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:

[1]Linux foundation: ”Xen tools”. Information available at: http://wiki.xen.org/wiki/Xen_tools

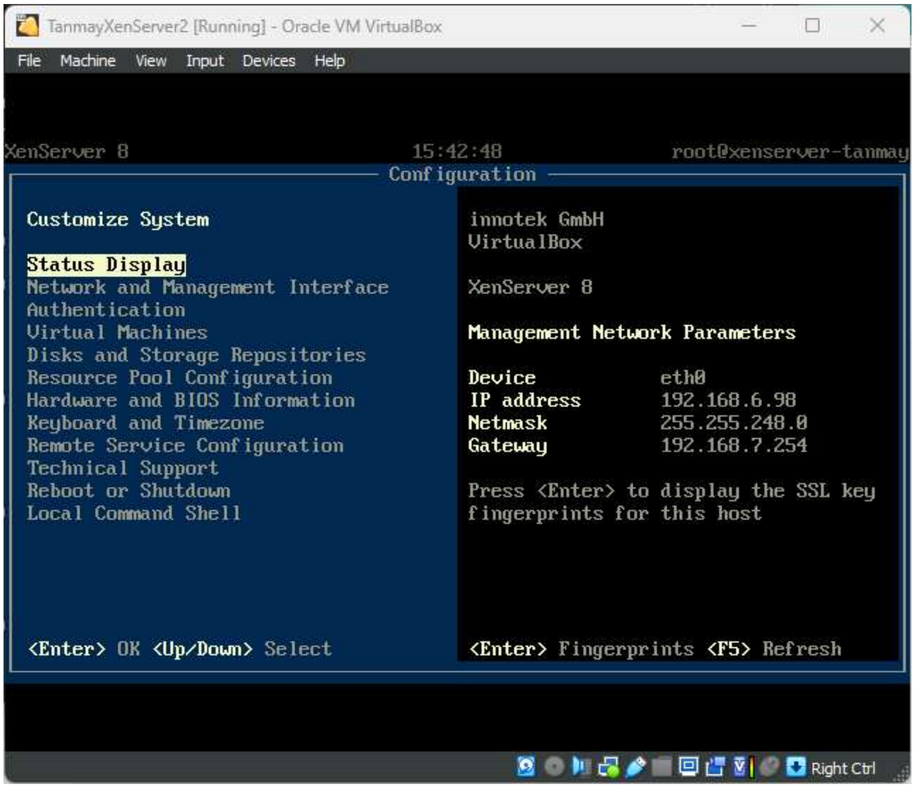
Last review: 16-03-2014: 13.51

[2] Bill HILL “Intro to Virtualization: Hardware, Software, Memory, Storage, Data and Network Virtualization Defined”. Information available at: http://www.petri.co.il/intro-tovirtualization.htm

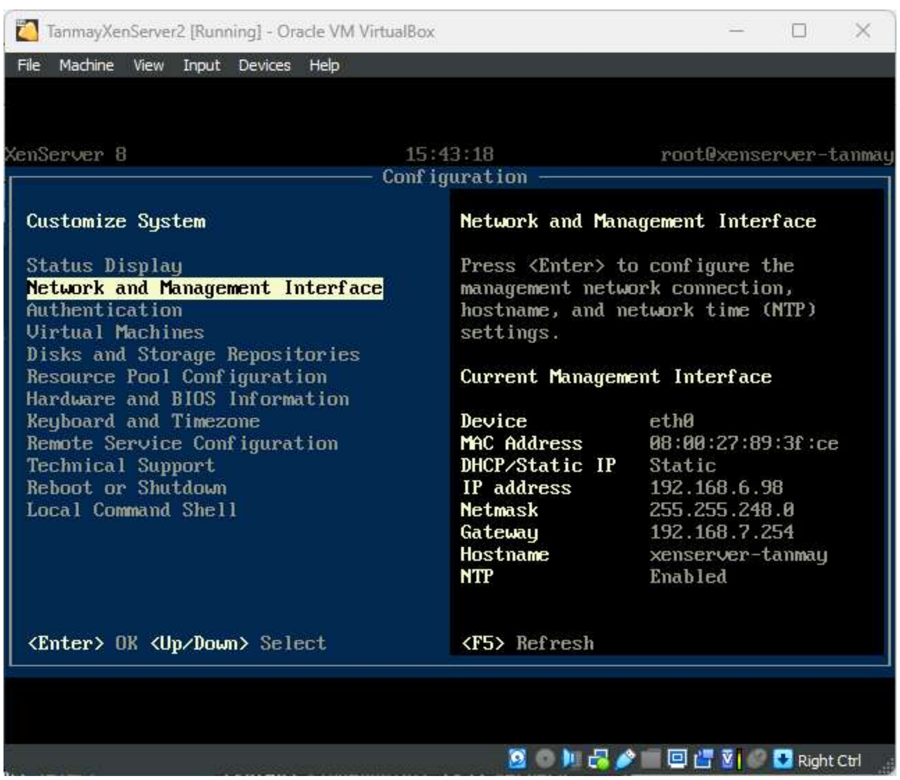
Last review: 16-03-2014: 12.55

[3] [Online]https://docs.citrix.com/en-us/citrix-hypervisor/install.html

[4] [Online]https://i12bretro.github.io/tutorials/0372. Html

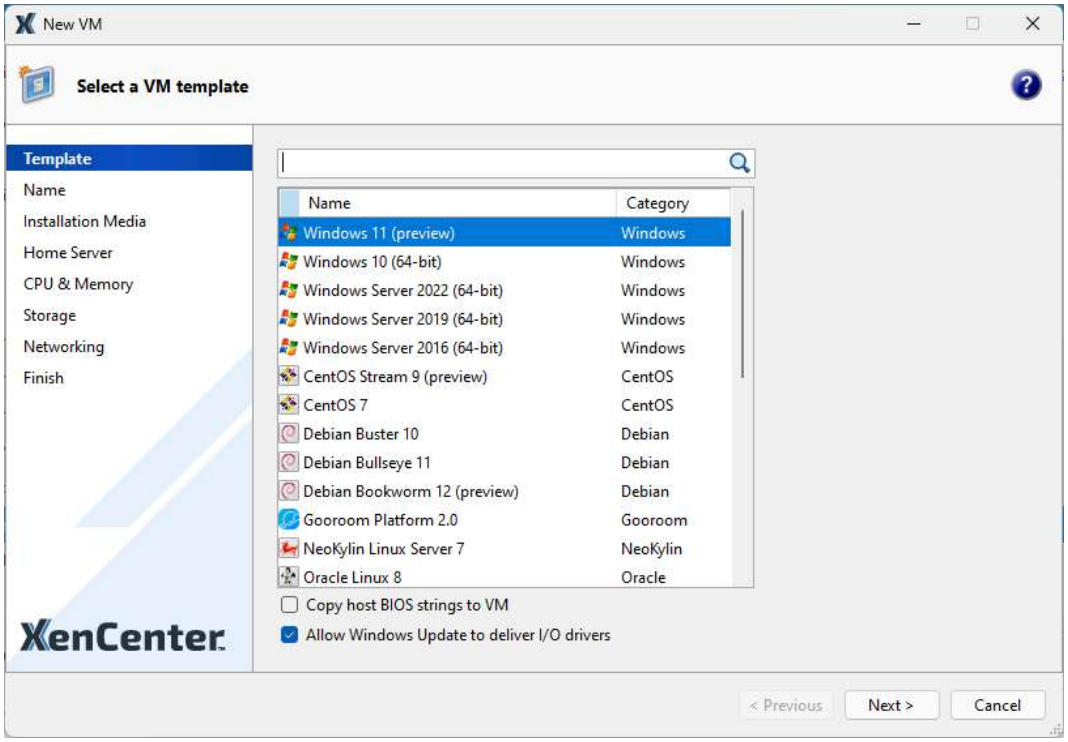


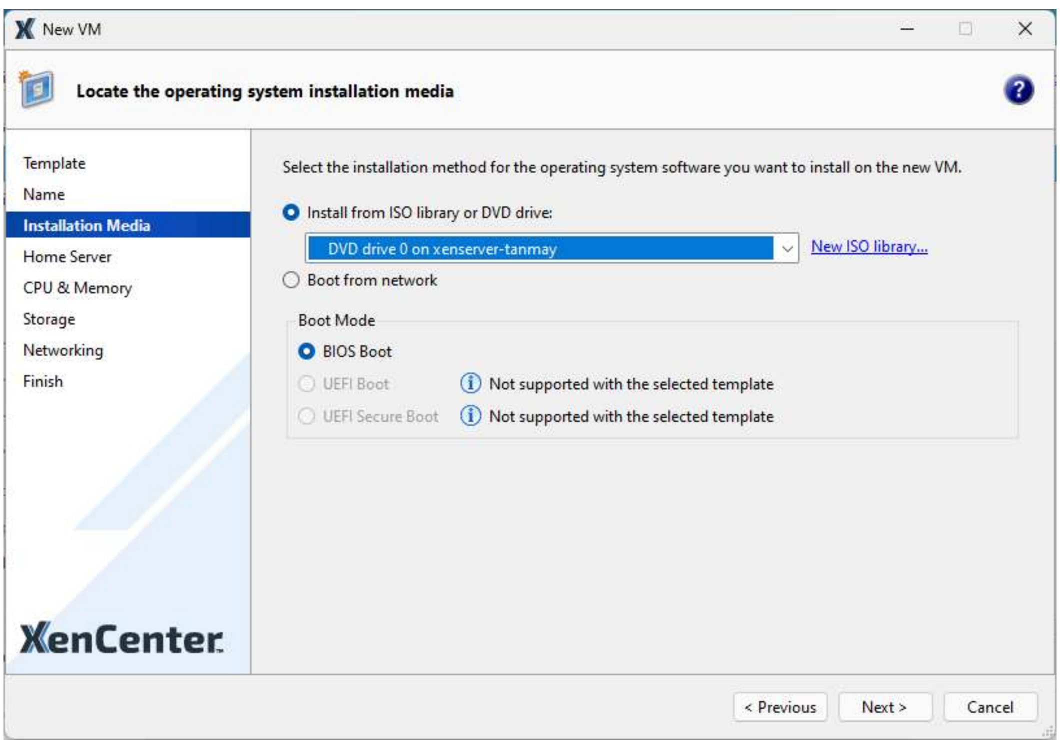
Steps 32-26

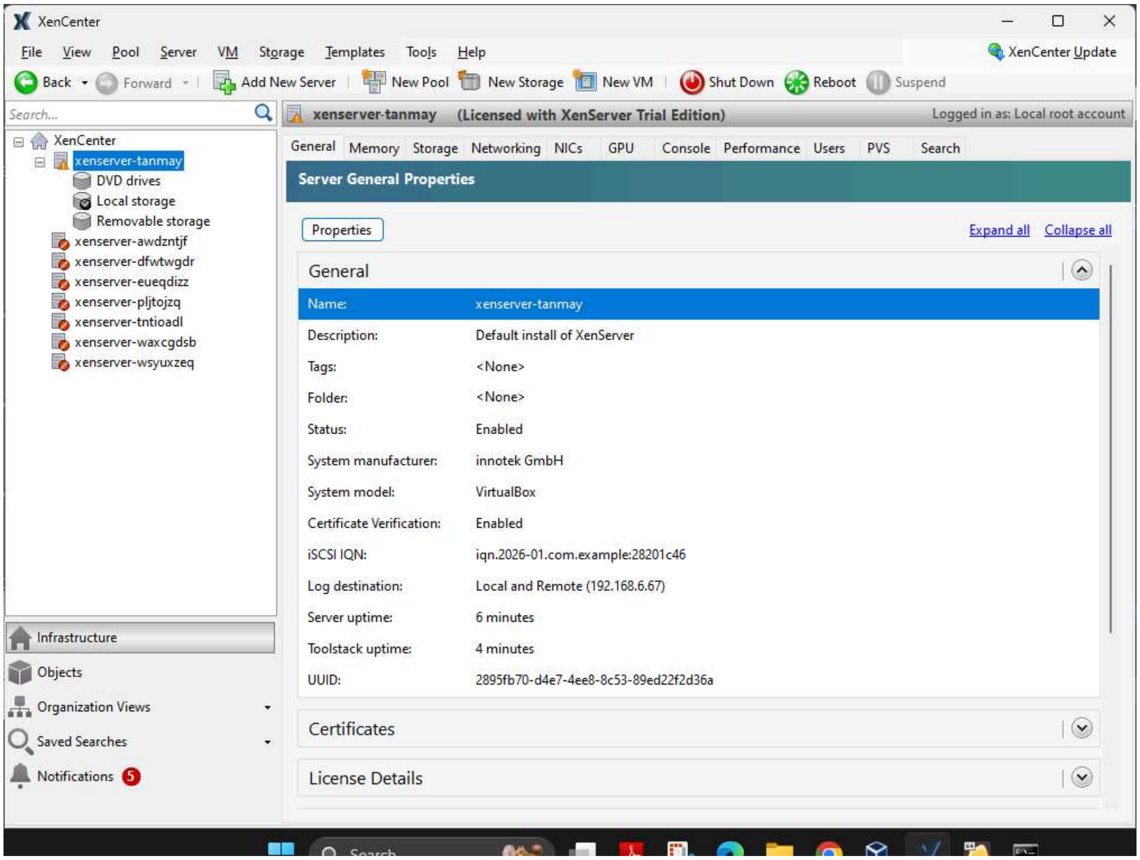


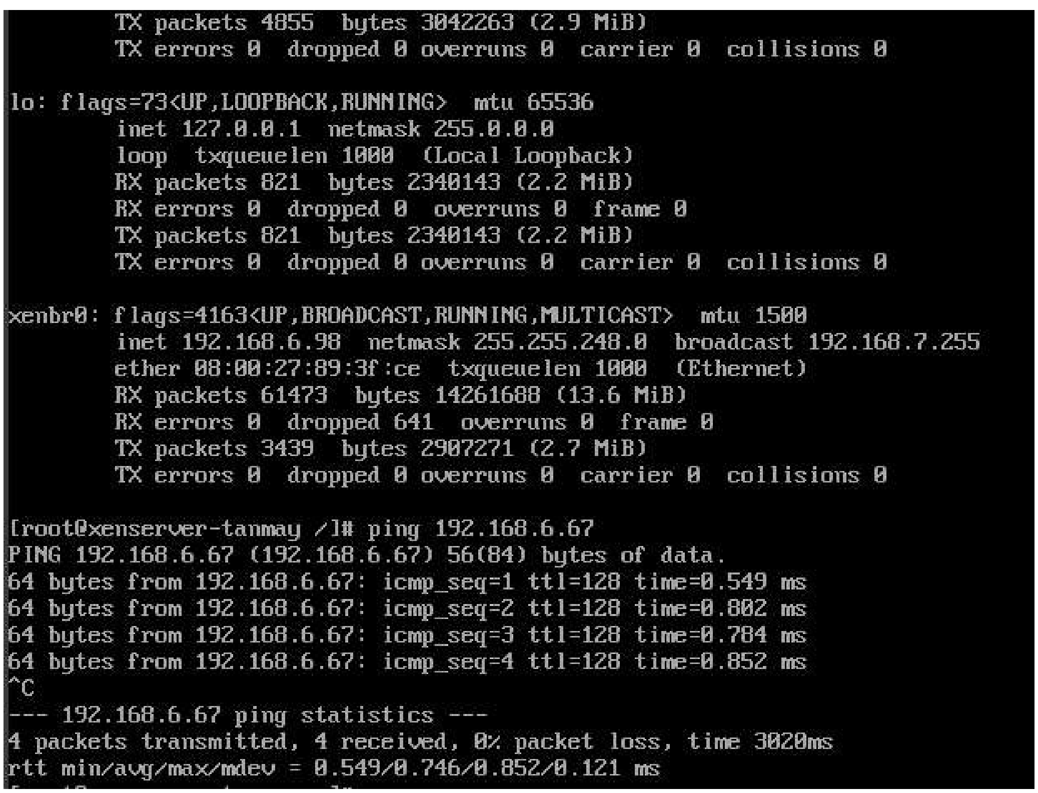
XenServer network and management interface

● **To connect XenCenter to the Citrix Hypervisor server:**

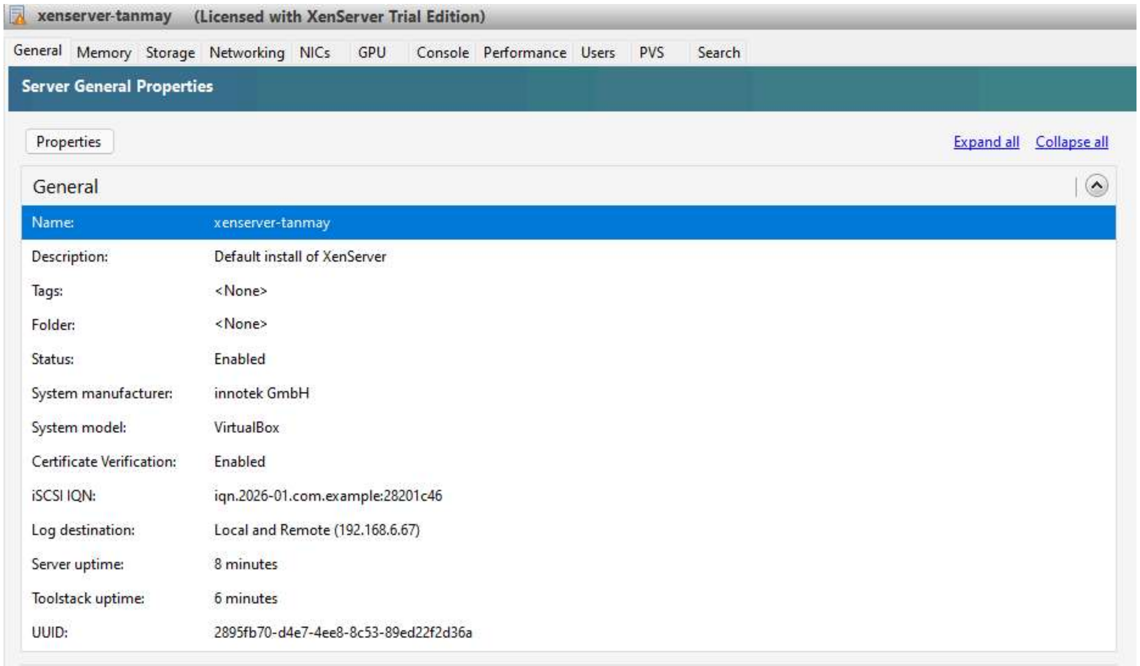








Ping information



XENSERVER running from xencenter